Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [MSCCH402]

FOURTH SEMESTER UNIVERSITY EXAMINATION, JUNE 2016

M.Sc. (BIOCHEMISTRY)

SPECTROSCOPY-II

Time : 03 Hours **Maximum Marks:80**

Instructions:
1. Questions 1 to 6 carry 15 marks each, answer any four.
2. Question 7 is compulsory, carries 20 marks.

1. **a.** Explain IR frequencies of alkanes and alkenes.

b. What is Fermi resonance in IR spectroscopy? Explain with an example.

c. How to analyse an IR spectrum? Explain in detail. **3+5+7**

2. Explain hypsochromic, bathochromic, hyperchromic and hypochromic shift in UV-Vis spectroscopy. **15**

3. **a.** Calculate the ESR frequency in a magnetic field of 25000 Gauss, of g=2 and $\beta=9.273 \times 10^{-24}\ JT^{-1}$

b. Draw the hyperfine energy levels diagram for CH_3 radical and predict the number of line in EPR spectra. **5+10**

4. **a.** Explain the principle of Mössbauer spectroscopy with appropriate illustration and list out its advantages and limitations.

b. Explain the basic principle involved in NQR and list out its applications. **10+5**

5. Explain the instrumentation of Photoelectron spectroscopy (PES) and its applications in detail. **15**

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6. **a.** Explain Auger effect.

b. Write a note on XRF and its applications.

c. Explain the instrumentation of XAS and its applications. **5+5+5**

7. **Answer any *TEN* of the following.** **10×2=20**

a. What are the differences between optical and electron microscopes?

b. Define fluorescence.

c. Explain in brief HOMO and LUMO.

d. Calculate the frequency of electromagnetic radiations corresponding to the following wavelength

   i. 2000 Å;

   ii. 4μm

e. What is finger print region in IR spectroscopy? Explain with an example.

f. Explain the effect of hydrogen bonding with an example in IR spectroscopy.

g. Only solid samples can be used in NQR spectroscopy. Justify.

h. Explain magnetic splitting in Mössbauer spectroscopy.

i. Predict the number of lines in ESR spectra for the following:

   i. $CH_3{}^{\cdot}CH_2$

   ii. $^{\cdot}CH_3$

j. Predict the number of lines in ESR spectra for the following:

   i. $C_6H_6^-$

   ii. V^{4+}

k. Define Photoelectric effect.

l. Define Koopman's theorem.